

Sampath Varma Datla

AI Engineer | LLM Systems & Agentic AI

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SUMMARY

AI Engineer with end-to-end ownership of LLM systems, RAG pipelines, and agentic AI in production on AWS. Delivered **four** AI and full-stack platforms across healthcare and ed-tech — cutting retrieval time by **70%**, scaling to **500+** concurrent users, and translating complex technical work into measurable business outcomes.

EDUCATION

GITAM University

Bachelor of Technology, Computer Science and Engineering

Sep. 2022 – Apr. 2026

EXPERIENCE

DATAi2i Private Limited

Junior AI Engineer | Promoted from AI Engineer Intern

May 2026 – Present

- Architected and deployed LLM-powered NLP inference pipelines for pharma clients on **AWS SageMaker** and **S3**, reducing deployment time by **30%** and streamlining **CI/CD** workflows.
- Fine-tuned and optimized LLMs (**LLM Fine-tuning**) for text classification, NER, and contextual search, achieving **25%** accuracy gain over baseline; pipeline observability tracked via **Langfuse**.

ZenithZap Beverages Private Limited

AI Engineer & Full-Stack Developer Intern

Aug. 2025 – Oct. 2025

- Delivered a production e-commerce platform using **React.js**, **Node.js**, and **Express.js**, growing MAUs past **5,000** and achieving a **30%** increase in brand visibility.
- Integrated a conversational AI chatbot via **LLM APIs**, boosting engagement by **4%**, cutting support queries by **25%**, and reducing latency by **40%** through **server-side caching**.

DATAi2i Private Limited

AI Engineer Intern

May 2025 – Jul. 2025

- Owned full SDLC of an internal knowledge assistant (**Gemini API**, **MongoDB**, **Streamlit**, **LlamaIndex**), from system design and data pipeline setup through production deployment.
- Accelerated document retrieval across **120+** files by **70%** via **semantic chunking** and **dense-vector indexing (RAG)**, achieving top-1 precision of **70%** and removing **60%** of off-topic results.

BMARG Innovative Solutions

Software Engineer Intern

Jan. 2025 – Apr. 2025

- Developed a **MERN**-stack clinic portal with **Razorpay** payment integration and **Firebase** Authentication, cutting onboarding time by **35%** and improving page-load speed by **40%**.
- Scaled to **500+** concurrent sessions by profiling and resolving critical **Node.js** scheduling bottlenecks with clinical staff.

PROJECTS

Enterprise Document Search Assistant | Python, Gemini API, MongoDB, Streamlit, RAG, Vector Search

2025

- Built a **RAG** pipeline with **dense-vector indexing** and **semantic segmentation (Python, Gemini API, MongoDB)**, cutting search time by **70%** and removing **60%** of off-topic results; monitored via **Langfuse** and **LangSmith**.

AI Fitness Coach | Python, LangChain, LangGraph, React.js, Node.js, OpenAI

2025

- Architected a multi-step agentic pipeline in **LangGraph** and **LangChain** (**OpenAI** APIs, **React.js** frontend), achieving **3×** faster plan generation via structured goal-parsing and synthesis stages.

LifePathBot — AI Student Success Platform | React, TypeScript, Vite, Tailwind CSS, Firebase

2025

- Built an **LLM**-backed platform with multi-file ingestion (**PDF, DOCX, PPTX**), goal tracking, and analytics dashboard using **React**, **TypeScript**, **Vite**, **Tailwind CSS**, and **Firebase**; piloted with **200+** students.

Exam Integrity Monitor | JavaScript, WebRTC, MediaPipe, face-api.js, Computer Vision

2024

- Engineered a real-time **30 FPS** webcam proctoring system using **MediaPipe**, **face-api.js**, and **WebRTC (JavaScript)**, reducing proctor workload by **60%** across **1,000+** exam sessions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

Generative AI & LLMs: LLMs, LLM Fine-tuning, RAG, Prompt Engineering, Vector Search, Embeddings, LangChain, LangGraph, LlamaIndex, CrewAI, n8n, Langfuse, LangSmith

ML & Computer Vision: Machine Learning, NLP, Computer Vision, TensorFlow, PyTorch, OpenCV, Scikit-learn, MLOps

Web & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, GraphQL, HTML5, CSS3, Tailwind CSS

Cloud, Databases & DevOps: AWS (SageMaker, S3), Docker, CI/CD, Firebase, MongoDB, MySQL, PostgreSQL, Supabase, Vercel

Tools & Version Control: Git, GitHub, Postman, Streamlit, VS Code, Claude Code, Agile

ACHIEVEMENTS

- 1st Place** — Internal Smart India Hackathon (2025), GITAM University; outranked **400+** competing teams
- Selected** — IIT Delhi Computer Graphics & CV Summer School; chosen from **1,500+** applicants
- Top 10** — IIT Kanpur Techkriti Technical Innovation Event
- President** — GITAM Aero Astro Club (2025–26); leading a STEM club of **50+** members across tech and non-tech domains
- Operations Lead** — IEEE CS GITAM Chapter (2024–25); organized workshops and cross-department initiatives